Theoretical foundations and learning

[06] Number theory for crypto (primes, factoring, modular arithmetic) — 5.47%

- Implementing Finite-Field / Modular Arithmetic — 64.71%
- Primes, Factoring, and Prime Generation — 23.53%
- Learning & Understanding Crypto Algorithms — 11.76%

[21] Conceptual crypto explanations and learning — 0.64%

- Where to start learning crypto (topics and references) — 50.00%
- Publishing venues and research quality (papers/conferences) — 50.00%

Cryptographic primitives and algorithms

[07] Public-key cryptography: encryption and digital signatures — 5.14%

- Public vs. private key roles (encrypt/decrypt semantics) — 43.75%
- Digital signatures: hashing, signing, and verification workflow — 37.50%
- Secure communication and key exchange over out-of-band channels — 12.50%

[08] Block cipher modes, IV/nonce, and padding — 4.18%

- Cryptographic security model and attack reasoning — 30.77%
- CTR/stream-style modes, nonces, and counters — 15.38%
- Encryption/decryption debugging in code (files/OpenSSL) — 15.38%

[14] Secure random number generation (entropy, seeding, PRNG) — 3.22%

- Generating random numbers & seeding pitfalls — 30.00%
- CSPRNG security model: state compromise & predictability — 20.00%
- True/HW randomness sources (TRNG) — 20.00%

[15] Cryptographic hash functions & collisions — 3.22%

- Cryptographic hash function properties (input/output, bit-length, preimage) — 60.00%
- Hash collisions in practice (files/strings, uniqueness) — 20.00%
- Hashing vs encryption (reversibility and keys) — 20.00%

Identity management and passwords

[04] Password hashing with salt (bcrypt, rainbow tables) — 7.40%

- Salts: per-user randomness and rainbow table defense — 56.52%
- Storing and verifying password hashes in a database (MD5-era patterns) — 39.13%
- Password-based encryption vs hashing (key derivation and data encryption) — 4.35%

[10] User authentication and account management — 3.86%

- Password hashing, salting, and account login management — 75.00%
- Server-side encryption and key/token storage — 25.00%

[13] Password strength and entropy estimation — 3.22%

- Passphrases vs complex passwords (dictionary words, usability, composition rules) — 60.00%
- Password search space from length and character set (brute-force cracking) — 30.00%
- Password/PIN entropy and guessability (bits of security) — 10.00%

Key infrastructure and trust (PKI)

[03] X.509 certificates, signing, and trust chains — 8.36%

- Private key & keystore management (PKCS#12, import/export/convert) — 19.23%
- Trust stores and certificate chain building/validation — 15.38%
- Debugging certificate/crypto verification code errors — 15.38%

[20] Trust, vendors, and real-world security risk — 1.93%

- Government backdoors, lawful access, and privacy debate — 33.33%
- Trust, proof, and digital signatures for claims — 16.67%
- Payment and credit card data handling security — 16.67%

Implementation and software development

[01] OpenSSL installation/build and runtime errors — 14.15%

- OpenSSL installation/build and linking (versions, packages, compilation) — 81.82%
- TLS/SSL runtime errors (certificates & connection issues) — 11.36%
- Key/certificate files and OpenSSL command usage — 6.82%

[05] Crypto library/API implementation in code — 5.47%

- Crypto library integration (OpenSSL, dependencies, builds) — 64.71%
- Password-based encryption parameters (salt, mode, padding) — 5.88%
- Hash algorithm choice and secure implementations — 5.88%

[18] Application data encryption & key storage — 2.57%

- Encryption/Decryption Code for Files and Strings — 75.00%
- Secure Key and Password Storage for Application Data — 25.00%

Communication and data handling

[02] Encoding and byte/string conversions (Base64, binary) — 11.90%

- Byte/string/character encoding conversions (Base64, binary, hashes) — 48.65%
- Password hashing implementation and verification (MD5 and checks) — 37.84%
- Symmetric encryption/decryption errors (keys, padding, file/text decoding) — 13.51%

[09] Client-server secure communication (HTTP/HTTPS, sessions) — 3.86%

- TLS/HTTPS Certificates and Trust Validation — 66.67%
- Secure Client–Server Authentication and Encrypted Password/Data Exchange — 33.33%

[11] File handling, integrity checks & checksums — 3.86%

- File checksums and hash verification (MD5, collisions) — 41.67%
- Digital signatures in PDF/documents — 16.67%
- Image steganography & JPEG data hiding — 16.67%

Security parameters and asset protection

[16] Key sizes and security parameters selection — 2.89%

- Key size, bit security level, and block/round parameters — 88.89%
- Key generation and encryption in application code (passwords, hashing, randomness) — 11.11%

[17] Key protection on devices (memory/disk access, threat model) — 2.89%

- Key storage in device memory (software vs hardware) — 55.56%
- Threat model & access control for encrypted data — 22.22%
- File/disk encryption with passwords (containers, boot, backups) — 22.22%

Security engineering and risk analysis

[12] Security vs performance (cost, speed, hardware limits) — 3.54%

- Performance optimization in crypto implementations (CPU/memory/hardware) — 54.55%
- Key sizes and computational feasibility of attacks (including quantum) — 36.36%
- Password hashing cost tuning (bcrypt/iterations/salt) against brute force — 9.09%

[19] Cryptosystem design pitfalls & attack resistance — 2.25%

- Designing crypto algorithms (and why they get broken) — 71.43%
- Protocol/implementation attack surface & standards — 28.57%